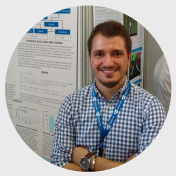
PAOLO VIVIANI

PhD – Research Engineer

Torino, ITALY paolo.vivi@gmail.com paoloviviani.github.io



RESEARCH ACTIVITY SUMMARY

I am a theoretical physicist turned computer scientist. I defended my PhD at University of Turin in 2019 and I am currently employed as research engineer at Noesis Solutions NV[1], a Belgian CAE software company. My main interests are High-Performance Computing and systems for Machine Learning and AI. I'm also interested in containers and cloud technologies applied to HPC.

Master's thesis

My research journey in HPC started in 2015: while drifting away from theoretical physics to more computational aspects, Professor Marco Aldinucci[2] proposed me a master's thesis on the parallelisation of a Fortran code that computed high-order analytical cross sections for interactions producing multiple Higgs' bosons [1]. The work revolved around isolating independent tasks and loop iterations in order to achieve the highest degree of parallelism, while also probing the suitability of different techniques and tools (e.g. FastFlow, OpenMP and also moving some kernels to the GPU). Despite the challenges related to legacy code, performance results fully satisfied expectations.

PhD activity

After the master's degree I won a six month scholarship at Computer Science department of University of Turin, funded by Noesis Solutions. The activities involved full time internship within the company to perform research on topics of common interest between my academic research group and the company itself. The same pattern applied to my PhD course, also funded by Noesis and advised by Marco Aldinucci, that started right after the scholarship.

The research topics I encountered spanned a broader scope than the typical PhD course, in fact I focused on both HPC and machine learning topics, first somehow separately, then the two tracks converged into my PhD thesis. Moreover, a large share of my time during the PhD has been dedicated to funded research projects: between 2015 and present day, I have been in charge of the technical aspects of six funded research project (from both EU funding and national funding). This task involved participating in project meetings, managing relations with international partners, presenting results at project reviews and writing deliverables. I also have some experience with the financial aspects of funded projects, as well as with proposal preparation and writing. The project I have been involved in are the following:

- MAssive Calculations on Hybrid systems (MACH) – ITEA2 project
- Fortissimo 2 – H2020-FoF project
- CloudFlow – FP7-I4MS project
- Blockchain for online Service Security – ICON national project
- VaProFam – ICON national project
- DC-CDS – ICON national project

Recurring topics are the convergence of cloud and HPC technologies, simulation workflows-as-a-service, knowledge-based optimization, and blockchain technologies for engineering. Some of these activities resulted in publications that I co-authored [7], [8].

During the PhD course I also attended the following schools, with topics ranging from machine learning and large graph mining, to performance evaluation and compiler optimization:

- Bertinoro International Spring School 2016
- Regularization methods in Machine Learning (RegML) 2016, Genova
- Advanced Computer Architecture and Compilation for High-Performance and Embedded Systems (ACACES) 2016, Fiuggi
- Advanced Course on Data Science & Machine Learning 2018, Siena

Finally, in 2018 and 2019 respectively, I supervised an industrial internship within Noesis, and I co-advised a master's thesis with Prof. Marco Aldinucci.

Research results

The first result was the development of a framework able to execute high performance linear algebra operations on CPU/GPU hybrid platforms without any expertise in parallel programming required to the user, while also meeting a number of industrial requirements, like being able to switch at runtime between CPU and GPU execution. This work resulted in a few publications [2]–[4], and it is now being used by the company in its product.

On the machine learning side, I worked on the develop-

[1] https://www.noesissolutions.com/
[2] http://alpha.di.unito.it/marco-aldinucci/

ment of new data modelling techniques to be included in Noesis' software package. One instance is the application of ensemble methods to regression/interpolation models in order to achieve better generalization accuracy and robustness when dealing with very sparse data. The results has been extremely satisfactory and such methodology is now featured in the product, despite no publications are available due to confidentiality.

Eventually, my interest moved to the opposite side of the data size spectrum: very large datasets. This required to move away from $O(n^3)$ models and, in this case, to explore deep neural networks. I developed a methodology and an implementation to let the user exploit DNNs without specific deep learning expertise (i.e. a simple speed/accuracy trade-off slider that is reflected to heuristics in the architecture definition).

The activity on deep neural networks drove me to explore the growing convergence between HPC and AI: in fact, large-scale distributed training of deep neural networks is the central topic of my PhD thesis [5]. The goal was two-fold: first to investigate theoretically the impact of different distributed training methodologies to the model convergence [6], and to propose a novel approach that could overcome possible limitations; then to develop a software prototype that could implement that approach efficiently at large scale. The work focused on de-centralised, fully asynchronous training, where back-propagation gradients are exchanged only between neighbours thanks to a communication stencil defined with a high-level API. This leveraged previous work on distributed memory models, and achieved marginally better performance when compared with state of the art frameworks. On the other hand, neighbours communications never proved to be beneficial to the overall convergence of the training. Despite this, the theoretical discussion of the link between scalability and convergence of deep learning models is still very relevant and I consider it as the highlight of my work.

Industrial contributions

Beside the research activity, during my time at Noesis, I contributed to the improvement of the internal state of practice for scientific computing, by introducing modern C++ tools for the development of numerical algorithm. I also re-designed from scratch the source code management process at company level.

At the moment, I am still working on machine learning-based surrogate models for engineering, while also following funded research projects and contributing to the improvement of our software's data analytics capabilities. As additional activities I also work on the improvement of internal processes, namely source code management, deployment strategies and continuous integration.

References

[1] P. Viviani, "Parallel computing techniques for high energy physics," Master's thesis, Physics Department, University of Torino, 2015.

[2] P. Viviani, M. Torquati, M. Aldinucci, and R. d'Ippolito, "Multiple back-end support for the armadillo linear algebra interface," in *In proc. of the 32nd ACM Symposium on Applied Computing (SAC)*, Marrakesh, Morocco, Apr. 2017, pp. 1566–1573.

[3] P. Viviani, M. Aldinucci, R. d'Ippolito, J. Lemeire, and D. Vucinic, "A flexible numerical framework for engineering—a response surface modelling application," in *Improved Performance of Materials: Design and Experimental Approaches*. Cham: Springer International Publishing, 2018, pp. 93–106.

[4] M. Drocco, P. Viviani, I. Colonnelli, M. Aldinucci, and M. Grangetto, "Accelerating spectral graph analysis through wavefronts of linear algebra operations," in *Proc. of 27th Euromicro Intl. Conference on Parallel Distributed and network-based Processing (PDP)*, Pavia, Italy: IEEE, 2019, pp. 9–16.

[5] P. Viviani, "Deep learning at scale with nearest neighbours communications," PhD thesis, Computer Science Department, University of Torino, Sep. 2019.

[6] P. Viviani, M. Drocco, D. Baccega, I. Colonnelli, and M. Aldinucci, "Deep learning at scale," in *Proc. of 27th Euromicro Intl. Conference on Parallel Distributed and network-based Processing (PDP)*, Pavia, Italy: IEEE, 2019, pp. 124–131.

[7] V. Reniers, D. Van Landuyt, P. Viviani, *et al.*, "Analysis of architectural variants for auditable blockchain-based private data sharing," in *Proceedings of the 34th ACM/SIGAPP Symposium on Applied Computing*, ser. SAC '19, Limassol, Cyprus: ACM, 2019, pp. 346–354.

[8] V. Reniers, Y. Gao, R. Zhang, *et al.*, "Authenticated and auditable data sharing via smart contract," in *Proceedings of the 35th ACM/SIGAPP Symposium on Applied Computing*, ser. SAC '20, Brno, Czech Republic: ACM, 2020, pp. 1–8.